

Task No.: 11**Date: 08-02-2026**

Title: Monitoring Wi-Fi Connectivity and Active Connection Details**Aim:**

To develop an Android Application that monitors Wi-Fi connectivity and active connection details.

Code:*MainActivity.kt*

```
package com.example.madtask11

import android.Manifest
import android.content.Context
import android.content.pm.PackageManager
import android.net.wifi.WifiManager
import android.os.Bundle
import android.text.format.Formatter
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.content.ContextCompat

class MainActivity : AppCompatActivity() {

    private lateinit var txtWifiInfo: TextView

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        txtWifiInfo = findViewById(R.id.txtWifiInfo)

        checkPermission()
```

```

}

private fun checkPermission() {

    if (ContextCompat.checkSelfPermission(
        this,
        Manifest.permission.ACCESS_FINE_LOCATION
    ) != PackageManager.PERMISSION_GRANTED
    ) {

        ActivityCompat.requestPermissions(
            this,
            arrayOf(Manifest.permission.ACCESS_FINE_LOCATION),
            100
        )

    } else {

        showWifiDetails()

    }
}

override fun onRequestPermissionsResult(
    requestCode: Int,
    permissions: Array<String>,
    grantResults: IntArray
) {

    super.onRequestPermissionsResult(
        requestCode,
        permissions,
        grantResults
    )

    if (requestCode == 100 &&
        grantResults.isNotEmpty() &&
        grantResults[0] == PackageManager.PERMISSION_GRANTED
    ) {

        showWifiDetails()
    }
}

```

```

    } else {

        txtWifiInfo.text = "Location Permission Required"

    }
}

```

```

private fun showWifiDetails() {

    val wifiManager =
        applicationContext.getSystemService(
            Context.WIFI_SERVICE
        ) as WifiManager

    if (!wifiManager.isWifiEnabled) {

        txtWifiInfo.text = "Wi-Fi Status: OFF"

        return
    }

    val wifiInfo = wifiManager.connectionInfo

    val ssid = wifiInfo.ssid
    val bssid = wifiInfo.bssid
    val ipAddress =
        Formatter.formatIpAddress(wifiInfo.ipAddress)
    val rssi = wifiInfo.rssi
    val linkSpeed = wifiInfo.linkSpeed

    val wifiDetails = ""

    Wi-Fi Status: ON

    Network SSID: $ssid

    BSSID: $bssid

    IP Address: $ipAddress

```

```

        Signal Strength: $rssi dBm
        Link Speed: $linkSpeed Mbps
        """".trimIndent()
        txtWifiInfo.text = wifiDetails
    }
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/txtTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name: Yoga Shri P K&#10;Roll No.: 23BCS192"
        android:textSize="22sp"
        android:textStyle="bold"
        android:gravity="center"/>

    <ScrollView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginTop="20dp">

        <TextView
            android:id="@+id/txtWifiInfo"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Wi-Fi details will appear here"
            android:textSize="18sp"
            android:padding="16dp"/>

    </ScrollView>
</LinearLayout>

```

Output:

Name: Yoga Shri P K
Roll No.: 23BCS192

Wi-Fi Status: ON

Network SSID:
"Airtel_Krishmoorthy"

BSSID: 58:07:f8:d6:d8:ea

IP Address: 192.168.1.6

Signal Strength: -52 dBm

Link Speed: 292 Mbps

Result:

Thus, Android Application that monitors wifi connectivity and active connection details is developed and executed successfully.